

Credit Card Approval Prediction

Phase 6: Project Testing

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. Model Evaluation Testing

All four models were evaluated on an identical held-out 20% test split (stratified to preserve the Approved/Rejected ratio), using Accuracy, Precision, Recall, F1-score, and a Confusion Matrix.

Model	Accuracy	F1-score	Result
Logistic Regression	0.902	0.944	Pass
Decision Tree	0.909	0.949	Pass
Random Forest (deployed)	0.939	0.966	Pass — Best
XGBoost	0.921	0.956	Pass

Acceptance criterion (NFR-3): F1-score ≥ 0.85 . All four models passed; Random Forest was selected for deployment.

2. Functional Test Cases

Test ID	Test Case	Expected Result	Status
TC-01	Load home page (GET /)	Form renders with all input fields empty	Pass
TC-02	Submit form with valid low-risk application	Return 'Approved' with confidence %	Pass
TC-03	Submit form with high EMI_PASTDUES (6)	Return 'Rejected' with confidence %	Pass
TC-04	Submit form with an unseen category value	Fallback to default encoder class, no crash	Pass
TC-05	Submit form with missing numeric field	Backend raises a clear 400 error, no silent failure	Pass

TC-06	Verify model.pkl loads correctly on Next step Next step ; model object is usable	Pass
TC-07	Verify prediction confidence is between 0.000000 and 1.000000 always in valid range	Pass
TC-08	Re-submit the same applicant profile and receive identical prediction both times (deterministic)	Pass

3. Data Validation Testing

- Verified application_record.csv and credit_record.csv merge correctly on the ID column with no row-count explosion.
- Verified no duplicate applicant rows remain after drop_duplicates().
- Verified DAYS_BIRTH and DAYS_EMPLOYED contain no negative values after abs() conversion.
- Verified STATUS_BIN and TARGET columns contain only 0/1 values.
- Verified the final TARGET distribution is realistically imbalanced (~87% Approved / 13% Rejected), not artificially perfect.

4. Known Issue Identified & Resolved

Issue: an early version of the target definition ("any single overdue month = Rejected") combined with the EMI_PASTDUES feature caused all four models to score 100% accuracy — a data leakage symptom, since EMI_PASTDUES was a near-perfect deterministic function of TARGET.

Resolution: the target definition was changed to require 2+ overdue months, plus ~6% realistic label noise to simulate manual underwriting overrides. Re-running the evaluation produced the believable, varied results shown in Section 1.